| INCH-POUND |

MIL-C-19978/18A
19 June 1990
SUPERSEDING
MIL-C-19978/18
20 OCTOBER 1972

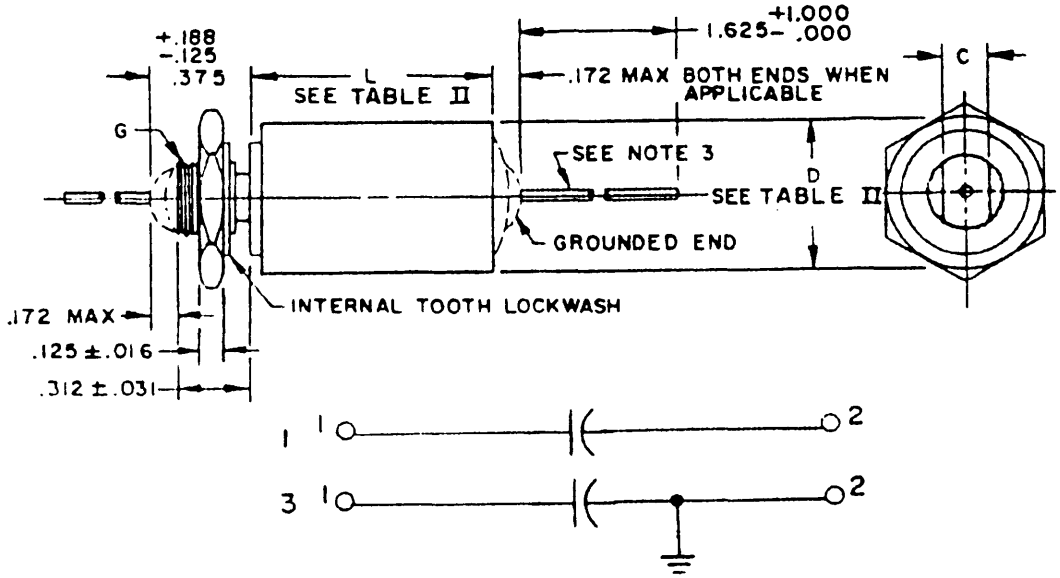
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, THREADED-STUD RETAINER, TUBULAR (UNINSULATED) (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CQR43

| INACTIVE FOR NEW EQUIPMENT |
| DESIGN AFTER 20 OCTOBER 1972. |

This specification is approved for use by all Dep rtments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-19978.



CIRCUIT DIAGRAM

FIGURE 1. Style CQR43 capacitors.

Ⓐ denotes changes.

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Inches	mm
.001	0.03
.004	0.10
.016	0.41
.031	0.79
.032	0.81
.125	3.18
.172	4.37
.188	4.80
.312	7.92
.375	9.54
1.000	25.40
1.625	41.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only.
3. Leads shall be of tinned solid wire, .032 +.004, -.001 (No. 20 AWG).
4. Mounting nuts and lockwashers shall be supplied.

FIGURE 1. Style CQR43 capacitors - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table II.

Case material: Nonmagnetic (end seal may be of magnetic material).

Dielectric material: Paper and polyethelene terephthalate (K).

Terminals: Axial-wire lead (see figure 1).

Rated voltage: See table II.

Rated temperature: -65°C to +125°C.

Capacitance (Cap.) (nom): See table II.

Capacitance tolerance: K, ±10 percent.

Dissipation factor (DF) (max): 1.0 percent.

Failure rate level: In accordance with MIL-C-19978.

Reliability: In accordance with MIL-C-19978.

Burn-in: In accordance with MIL-C-19978.

Radiographic inspection: In accordance with MIL-C-19978.

Seal: Method 112 of MIL-STD-202, test condition letter A.

Barometric pressure: In accordance with MIL-C-19978.

Test points (see figure 1, circuit diagram):

Circuit diagram 1: Between terminals and case.

Circuit diagram 3: Between ungrounded terminal and case.

Insulation resistance (IR):

Terminal to terminal: See table I.

TABLE I. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	
	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminal to case: Greater than 10,000 megohms.

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Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:

Direction and duration of motion: 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter D.

Stability at low and high termparature:

Low temperature:

Test temperature: -65°C +0°C, -5°C.

Capacitance change (max): -10 percent.

High temperature:

Test temperature: +125°C +5°C, -0°C.

Capacitance change (max): ±10 percent.

Life:

Capacitance change (max): ±5 percent.

Resistance to soldering heat: In accordance with MIL-C-19978.

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Ⓐ TABLE II. Style CQR43 capacitors.

Part or Identifying Number (PIN) 1/	DC voltage rating	Capacitance rating	Failure rate level	Case dimensions			
				Characteristic K			
				Circuit 1		Circuit 3	
				L±.031	D +.015 -.005	L±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQR43A-KB822K3-	100	.0082	L, M, P, R, S	.771	.400	.708	.400
CQR43A-KB103K3-	100	.010	"	.771	.400	.708	.400
CQR43A-KB123K3-	100	.012	"	.771	.400	.708	.400
CQR43A-KB153K3-	100	.015	"	.771	.400	.708	.400
CQR43A-KB393K3-	100	.039	"	.896	.400	.833	.400
CQR43A-KB473K3-	100	.047	"	.896	.400	.833	.400
CQR43A-KB823K3-	100	.082	"	.896	.400	.833	.400
CQR43A-KB104K3-	100	.10	"	.896	.400	.833	.400
CQR43A-KB124K3-	100	.12	"	1.146	.400	1.083	.400
CQR43A-KB154K3-	100	.15	"	1.146	.400	1.083	.400
CQR43A-KB184K3-	100	.18	"	1.396	.400	1.333	.400
CQR43A-KB224K3-	100	.22	"	1.396	.400	1.333	.400
CQR43A-KB274K3-	100	.27	"	1.146	.562	1.083	.562
CQR43A-KB334K3-	100	.33	"	1.146	.562	1.083	.562
CQR43A-KB394K3-	100	.39	"	1.406	.562	1.344	.562
CQR43A-KB474K3-	100	.47	"	1.406	.562	1.344	.562
CQR43A-KB564K3-	100	.56	"	1.656	.562	1.594	.562
CQR43A-KB684K3-	100	.68	"	1.656	.562	1.594	.562
CQR43A-KB824K3-	100	.82	"	1.656	.670	1.594	.670
CQR43A-KB105K3-	100	1.0	"	1.656	.670	1.594	.670
CQR43A-KC392K3-	200	.0039	"	.771	.400	.708	.400
CQR43A-KC472K3-	200	.0047	"	.771	.400	.708	.400
CQR43A-KC562K3-	200	.0056	"	.771	.400	.708	.400
CQR43A-KC682K3-	200	.0068	"	.771	.400	.708	.400
CQR43A-KC183K3-	200	.018	"	.896	.400	.833	.400
CQR43A-KC223K3-	200	.022	"	.896	.400	.833	.400
CQR43A-KC273K3-	200	.027	"	.896	.400	.833	.400
CQR43A-KC333K3-	200	.033	"	.896	.400	.833	.400
CQR43A-KC393K3-	200	.039	"	.896	.400	.833	.400
CQR43A-KC473K3-	200	.047	"	.896	.400	.833	.400
CQR43A-KC563K3-	200	.056	"	.896	.400	.833	.400
CQR43A-KC683K3-	200	.068	"	.896	.400	.833	.400
CQR43A-KC823K3-	200	.082	"	1.146	.400	1.083	.400
CQR43A-KC104K3-	200	.10	"	1.146	.400	1.083	.400
CQR43A-KC124K3-	200	.12	"	1.396	.400	1.333	.400
CQR43A-KC154K3-	200	.15	"	1.396	.400	1.333	.400
CQR43A-KC184K3-	200	.18	"	1.146	.562	1.083	.562
CQR43A-KC224K3-	200	.22	"	1.146	.562	1.083	.562
CQR43A-KC274K3-	200	.27	"	1.406	.562	1.344	.562
CQR43A-KC334K3-	200	.33	"	1.406	.562	1.344	.562
CQR43A-KC394K3-	200	.39	"	1.656	.562	1.594	.562
CQR43A-KC474K3-	200	.47	"	1.656	.562	1.594	.562
CQR43A-KC564K3-	200	.56	"	1.656	.670	1.594	.670
CQR43A-KC684K3-	200	.68	"	1.656	.670	1.594	.670
CQR43A-KC824K3-	200	.82	"	2.156	.750	2.094	.750
CQR43A-KC105K3-	200	1.0	"	2.156	.750	2.094	.750

See footnote at end of table.

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Ⓐ TABLE II. Style CQR43 capacitors - Continued.

Part or Identifying Number (PIN) 1/	DC voltage rating	Capacitance rating	Failure rate level	Case dimensions			
				Characteristic K			
				Circuit 1		Circuit 3	
				L±.031	$D^{+.015}_{-.005}$	L±.031	$D^{+.015}_{-.005}$
	Volts	µF		Inches	Inches	Inches	Inches
CQR43A-KE272K3-	400	.0027	L, M, P, R, S	.771	.400	.708	.400
CQR43A-KE332K3-	400	.0033	"	.771	.400	.708	.400
CQR43A-KE123K3-	400	.012	"	.896	.400	.833	.400
CQR43A-KE153K3-	400	.015	"	.896	.400	.833	.400
CQR43A-KE273K3-	400	.027	"	.896	.400	.833	.400
CQR43A-KE333K3-	400	.033	"	.896	.400	.833	.400
CQR43A-KE393K3-	400	.039	"	1.146	.400	1.083	.400
CQR43A-KE473K3-	400	.047	"	1.146	.400	1.083	.400
CQR43A-KE563K3-	400	.056	"	1.396	.400	1.333	.400
CQR43A-KE683K3-	400	.068	"	1.396	.400	1.333	.400
CQR43A-KE823K3-	400	.082	"	1.146	.562	1.083	.562
CQR43A-KE104K3-	400	.10	"	1.146	.562	1.083	.562
CQR43A-KE124K3-	400	.12	"	1.406	.562	1.344	.562
CQR43A-KE154K3-	400	.15	"	1.406	.562	1.344	.562
CQR43A-KE184K3-	400	.18	"	1.656	.562	1.594	.562
CQR43A-KE224K3-	400	.22	"	1.656	.562	1.594	.562
CQR43A-KE274K3-	400	.27	"	1.656	.670	1.594	.670
CQR43A-KE334K3-	400	.33	"	1.656	.670	1.594	.670
CQR43A-KE394K3-	400	.39	"	2.156	.750	2.094	.750
CQR43A-KE474K3-	400	.47	"	2.156	.750	2.094	.750
CQR43A-KF102K3-	600	.0010	"	.771	.400	.708	.400
CQR43A-KF122K3-	600	.0012	"	.771	.400	.708	.400
CQR43A-KF152K3-	600	.0015	"	.771	.400	.708	.400
CQR43A-KF182K3-	600	.0018	"	.771	.400	.708	.400
CQR43A-KF222K3-	600	.0022	"	.771	.400	.708	.400
CQR43A-KF272K3-	600	.0027	"	.896	.400	.833	.400
CQR43A-KF332K3-	600	.0033	"	.896	.400	.833	.400
CQR43A-KF392K3-	600	.0039	"	.896	.400	.833	.400
CQR43A-KF472K3-	600	.0047	"	.896	.400	.833	.400
CQR43A-KF562K3-	600	.0056	"	.896	.400	.833	.400
CQR43A-KF682K3-	600	.0068	"	.896	.400	.833	.400
CQR43A-KF822K3-	600	.0082	"	.896	.400	.833	.400
CQR43A-KF103K3-	600	.010	"	.896	.400	.833	.400
CQR43A-KF123K3-	600	.012	"	.896	.400	.833	.400
CQR43A-KF153K3-	600	.015	"	.896	.400	.833	.400
CQR43A-KF183K3-	600	.018	"	.896	.400	.833	.400
CQR43A-KF223K3-	600	.022	"	.896	.400	.833	.400
CQR43A-KF273K3-	600	.027	"	1.146	.400	1.083	.400
CQR43A-KF333K3-	600	.033	"	1.146	.400	1.083	.400
CQR43A-KF393K3-	600	.039	"	1.396	.400	1.333	.400
CQR43A-KF473K3-	600	.047	"	1.396	.400	1.333	.400
CQR43A-KF563K3-	600	.056	"	1.146	.562	1.083	.562
CQR43A-KF683K3-	600	.068	"	1.146	.562	1.083	.562
CQR43A-KF823K3-	600	.082	"	1.406	.562	1.344	.562
CQR43A-KF104K3-	600	.10	"	1.406	.562	1.344	.562

See footnote at end of table.

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Ⓐ TABLE II. Style CQR43 capacitors - Continued.

Part or Identifying Number (PIN) 1/	DC voltage rating	Capacitance rating	Failure rate level	Case dimensions			
				Characteristic K			
				Circuit 1		Circuit 3	
				L±.031	D +.015 -.005	L±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQR43A-KF124K3-	600	.12	L, M, P, R, S	1.656	.562	1.594	.562
CQR43A-KF154K3-	600	.15	"	1.656	.562	1.594	.562
CQR43A-KF184K3-	600	.18	"	1.656	.670	1.594	.670
CQR43A-KF224K3-	600	.22	"	1.656	.670	1.594	.670
CQR43A-KF274K3-	600	.27	"	2.156	.750	2.094	.750
CQR43A-KF334K3-	600	.33	"	2.156	.750	2.094	.750
CQR43A-KF394K3-	600	.39	"	2.406	.750	2.344	.750
CQR43A-KF474K3-	600	.47	"	2.406	.750	2.344	.750

1/ Complete PIN shall include additional symbols to indicate circuit and failure rate level, as applicable. The term Part or Identifying Number (PIN) is equivalent to the term part number which was previously used in the specification.

TABLE III. Millimeter equivalents to decimal inches for table II.

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.005	0.13	.562	14.27	.833	21.16	1.344	34.14	2.094	53.19
.015	0.38	.670	17.02	.896	22.76	1.396	35.46	2.156	54.76
.031	0.79	.708	17.98	1.083	27.51	1.406	35.71	2.344	59.54
.068	1.73	.750	19.05	1.146	29.11	1.594	40.49	2.406	61.11
.400	10.16	.771	19.58	1.333	33.86	1.656	42.06		

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NOTE: Style CQR43 capacitors are substitutable for style CP13 capacitors formerly covered in MIL-C-25/1K as shown in table IV.

TABLE IV. Substitutability data.

From	To	From	To
MIL-C-25/1K (CP13)	MIL-C-19978/18 (CQR43)	MIL-C-25/1K (CP13)	MIL-C-19978/18 (CQR43)
CP13A-KB822K-	CQR43A-KB822K3-	CP13A-KE273K-	CQR43A-KE273K3-
CP13A-KB103K-	CQR43A-KB103K3-	CP13A-KE333K-	CQR43A-KE333K3-
CP13A-KB123K-	CQR43A-KB123K3-	CP13A-KE393K-	CQR43A-KE393K3-
CP13A-KB153K-	CQR43A-KB153K3-	CP13A-KE473K-	CQR43A-KE473K3-
CP13A-KB393K-	CQR43A-KB393K3-	CP13A-KE563K-	CQR43A-KE563K3-
CP13A-KB473K-	CQR43A-KB473K3-	CP13A-KE683K-	CQR43A-KE683K3-
CP13A-KB823K-	CQR43A-KB823K3-	CP13A-KE823K-	CQR43A-KE823K3-
CP13A-KB104K-	CQR43A-KB104K3-	CP13A-KE104K-	CQR43A-KE104K3-
CP13A-KB124K-	CQR43A-KB124K3-	CP13A-KE124K-	CQR43A-KE124K3-
CP13A-KB154K-	CQR43A-KB154K3-	CP13A-KE154K-	CQR43A-KE154K3-
CP13A-KB184K-	CQR43A-KB184K3-	CP13A-KE184K-	CQR43A-KE184K3-
CP13A-KB224K-	CQR43A-KB224K3-	CP13A-KE224K-	CQR43A-KE224K3-
CP13A-KB274K-	CQR43A-KB274K3-	CP13A-KE274K-	CQR43A-KE274K3-
CP13A-KB334K-	CQR43A-KB334K3-	CP13A-KE334K-	CQR43A-KE334K3-
CP13A-KB394K-	CQR43A-KB394K3-	CP13A-KE394K-	CQR43A-KE394K3-
CP13A-KB474K-	CQR43A-KB474K3-	CP13A-KE474K-	CQR43A-KE474K3-
CP13A-KB564K-	CQR43A-KB564K3-	CP13A-KF102K-	CQR43A-KF102K3-
CP13A-KB684K-	CQR43A-KB684K3-	CP13A-KF122K-	CQR43A-KF122K3-
CP13A-KB824K-	CQR43A-KB824K3-	CP13A-KF152K-	CQR43A-KF152K3-
CP13A-KB105K-	CQR43A-KB105K3-	CP13A-KF182K-	CQR43A-KF182K3-
CP13A-KC392K-	CQR43A-KC392K3-	CP13A-KF222K-	CQR43A-KF222K3-
CP13A-KC472K-	CQR43A-KC472K3-	CP13A-KF272K-	CQR43A-KF272K3-
CP13A-KC562K-	CQR43A-KC562K3-	CP13A-KF332K-	CQR43A-KF332K3-
CP13A-KC682K-	CQR43A-KC682K3-	CP13A-KF392K-	CQR43A-KF392K3-
CP13A-KC183K-	CQR43A-KC183K3-	CP13A-KF472K-	CQR43A-KF472K3-
CP13A-KC223K-	CQR43A-KC223K3-	CP13A-KF562K-	CQR43A-KF562K3-
CP13A-KC273K-	CQR43A-KC273K3-	CP13A-KF682K-	CQR43A-KF682K3-
CP13A-KC333K-	CQR43A-KC333K3-	CP13A-KF822K-	CQR43A-KF822K3-
CP13A-KC393K-	CQR43A-KC393K3-	CP13A-KF103K-	CQR43A-KF103K3-
CP13A-KC473K-	CQR43A-KC473K3-	CP13A-KF123K-	CQR43A-KF123K3-
CP13A-KC563K-	CQR43A-KC563K3-	CP13A-KF153K-	CQR43A-KF153K3-
CP13A-KC683K-	CQR43A-KC683K3-	CP13A-KF183K-	CQR43A-KF183K3-
CP13A-KC823K-	CQR43A-KC823K3-	CP13A-KF223K-	CQR43A-KF223K3-
CP13A-KC104K-	CQR43A-KC104K3-	CP13A-KF273K-	CQR43A-KF273K3-
CP13A-KC124K-	CQR43A-KC124K3-	CP13A-KF333K-	CQR43A-KF333K3-
CP13A-KC154K-	CQR43A-KC154K3-	CP13A-KF393K-	CQR43A-KF393K3-
CP13A-KC184K-	CQR43A-KC184K3-	CP13A-KF473K-	CQR43A-KF473K3-
CP13A-KC224K-	CQR43A-KC224K3-	CP13A-KF563K-	CQR43A-KF563K3-
CP13A-KC274K-	CQR43A-KC274K3-	CP13A-KF683K-	CQR43A-KF683K3-
CP13A-KC334K-	CQR43A-KC334K3-	CP13A-KF823K-	CQR43A-KF823K3-
CP13A-KC394K-	CQR43A-KC394K3-	CP13A-KF104K-	CQR43A-KF104K3-
CP13A-KC474K-	CQR43A-KC474K3-	CP13A-KF124K-	CQR43A-KF124K3-
CP13A-KC564K-	CQR43A-KC564K3-	CP13A-KF154K-	CQR43A-KF154K3-
CP13A-KC684K-	CQR42A-KC684K3-	CP13A-KF184K-	CQR43A-KF184K3-
CP13A-KC824K-	CQR43A-KC824K3-	CP13A-KF224K-	CQR43A-KF224K3-
CP13A-KC105K-	CQR43A-KC105K3-	CP13A-KF274K-	CQR43A-KF274K3-
CP13A-KE272K-	CQR43A-KE272K3-	CP13A-KF334K-	CQR43A-KF334K3-
CP13A-KE332K-	CQR43A-KE332K3-	CP13A-KF394K-	CQR43A-KF394K3-
CP13A-KE123K-	CQR43A-KE123K3-	CP13A-KF474K-	CQR43A-KG474K3-
CP13A-KE153K-	CQR43A-KE153K3-		

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC, OS, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5910-1698)

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